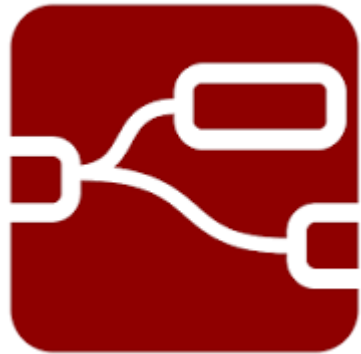




Node – RED

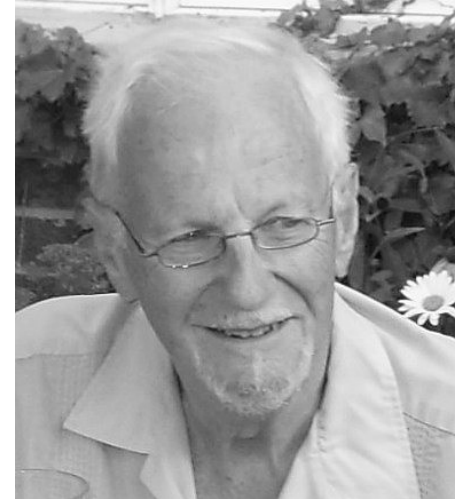
Low-code programming for event-driven applications

Node-RED is a programming tool for wiring together hardware devices, APIs and online services in new and interesting ways.

The screenshot shows the Node-RED website homepage. At the top is a dark navigation bar with the 'Node-RED' logo on the left and links for 'home', 'about', 'blog', 'documentation', 'forum', 'flows', and 'github' on the right. The 'home' link is underlined. Below the navigation bar is a large red hero section. In the center of this section, the text 'Node-RED' is displayed in a large, white, serif font. Below it, in a smaller white sans-serif font, is the tagline 'Low-code programming for event-driven applications'. Further down, in an even smaller white font, is 'Latest version: v1.0.6 (npm)'. Below the hero section is a light gray area. On the left side of this area, there is a paragraph of text: 'Node-RED is a programming tool for wiring together hardware devices, APIs and online services in new and interesting ways. It provides a browser-based editor that makes it easy to wire together flows using the wide range of nodes in the palette that can be deployed to its runtime in a single-click.' Below this text are three buttons: 'Features', 'Get Started', and 'Community'. On the right side of this gray area is a video player. The video player has a title 'Node-RED Introduction' and a large play button in the center. The video frame shows a screenshot of the Node-RED web interface with various nodes and a flow diagram.



- Paradigma basado en flujos.
“Flow-based programming” (FBP)
by Paul Morrison (60’)
- Proceso de **cajas negras** independientes
conectadas por **paquetes de informacion**
- Funciones independientes,
intercambiables, configurables
- El mundo es asíncrono
- Actualidad: Pure-Functions, Inmutability



J. Paul Morrison

author de “*Flow-Based
Programming: A New Approach to
Application Development*”

<https://jpaulm.github.io>





- Proyecto de OpenJS por IBM
- Licencia Apache 2
- Github <https://github.com/node-red>

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- Node.js
- Orientado a eventos
- Asíncrono
- Multipropósito
- JSON -WEBSOCKET – API - MQTT



 Code Wrangling

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OSS Developer ·
@node-red

👤 @IBM

✉ [nick.oleary@](mailto:nick.oleary@ibm.com)



Dave Conway-Jones

dceejay

Follow

👤 IBM UK labs

📍 Winchester, UK

🌐 <http://nodered.org>



Quick Start

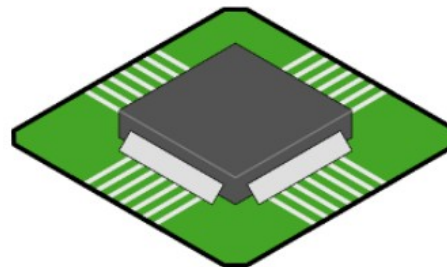
Check out <http://nodered.org/docs/getting-started/> for full instructions on getting started.

1. `sudo npm install -g --unsafe-perm node-red`
2. `node-red`
3. Open <http://localhost:1880>



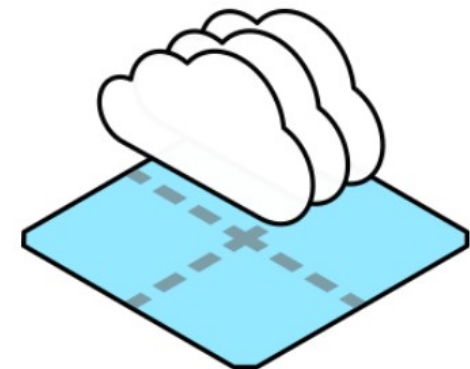
Run locally

- Getting started
- Docker



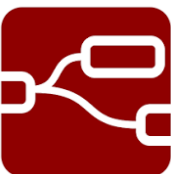
On a device

- Raspberry Pi
- BeagleBone Black
- Interacting with Arduino
- Android



In the cloud

- IBM Cloud
- SenseTecnica FRED
- Amazon Web Services
- Microsoft Azure



Node-RED

Deploy

filter nodes

input

inject

catch

mqtt

http

websocket

tcp

udp

serial

output

function

social

storage

analysis

advanced

Sheet 1

Node-RED GitHub

Bluemix monitor

Slack Bot

+

Home Energy

connected

Filter dupes

{

msg.payload

Node-RED GitHub Hooks

connected

/home/knolleary/github_hooks.json

timestamp

msg.payload

info

debug

Node

Name	SlackHook
Type	http in
ID	40c91d4d.bf36e4

Properties

Provides an input node for http requests, allowing the creation of simple web services.

The resulting message has the following properties:

- msg.req : [http request](#)
- msg.res : [http response](#)

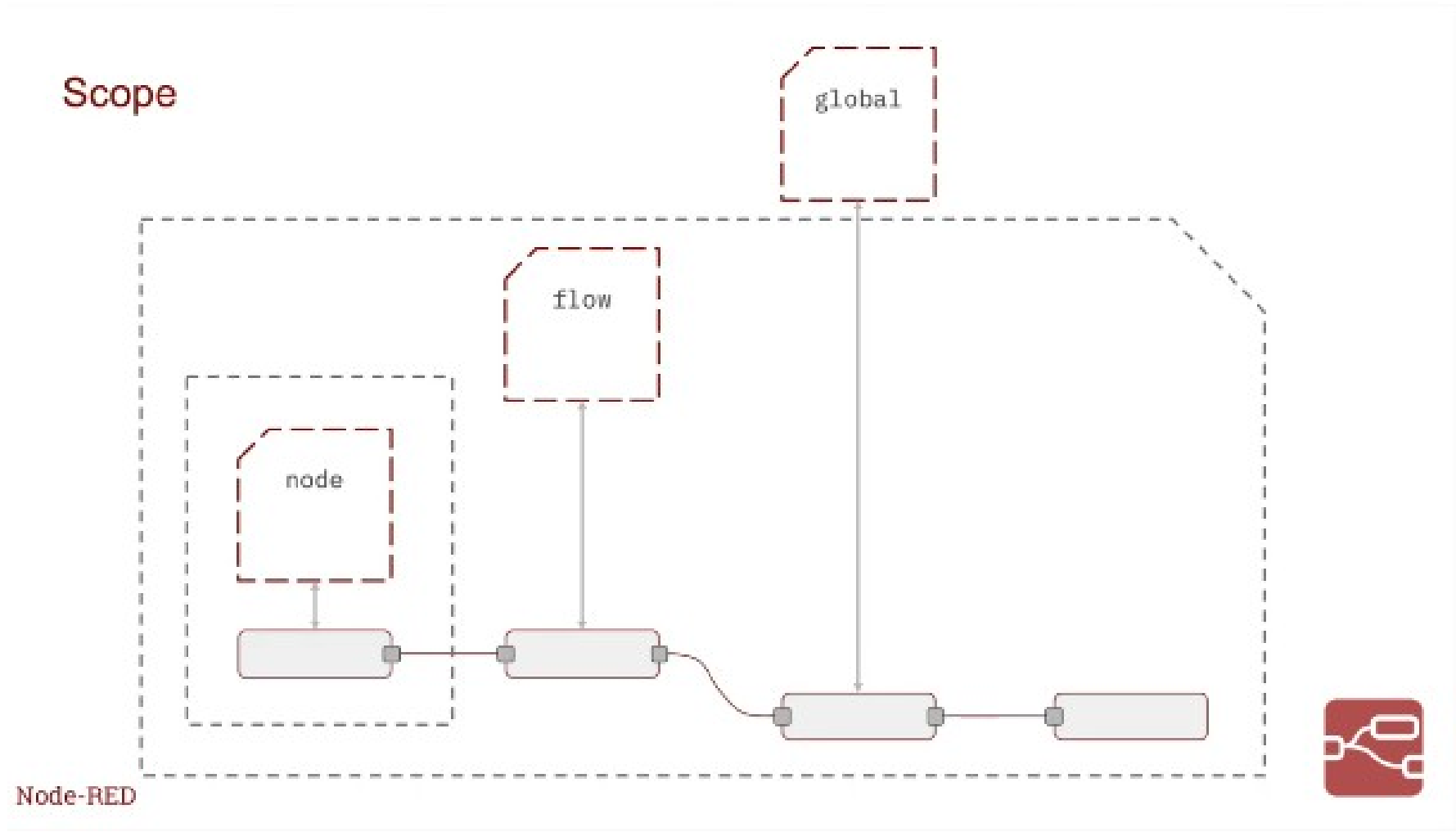
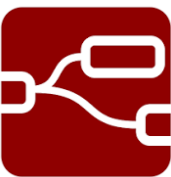
For POST/PUT requests, the body is available under `msg.req.body`. This uses the [Express bodyParser middleware](#) to parse the content to a JSON object.

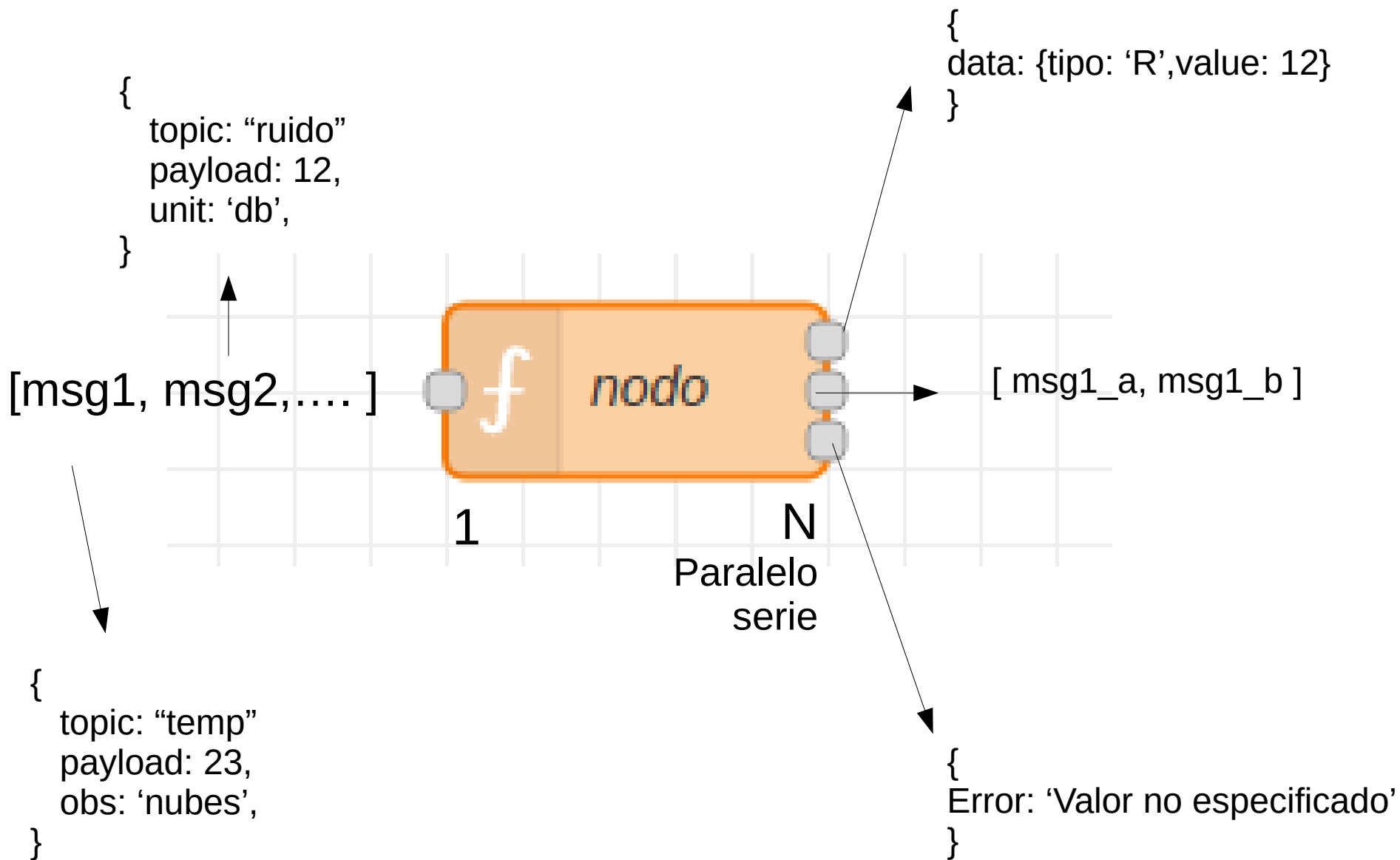
By default, this expects the body of the request to be url encoded:

foo=bar&this=that

To send JSON encoded data to the node, the content-type header of the request must be set to `application/json`.

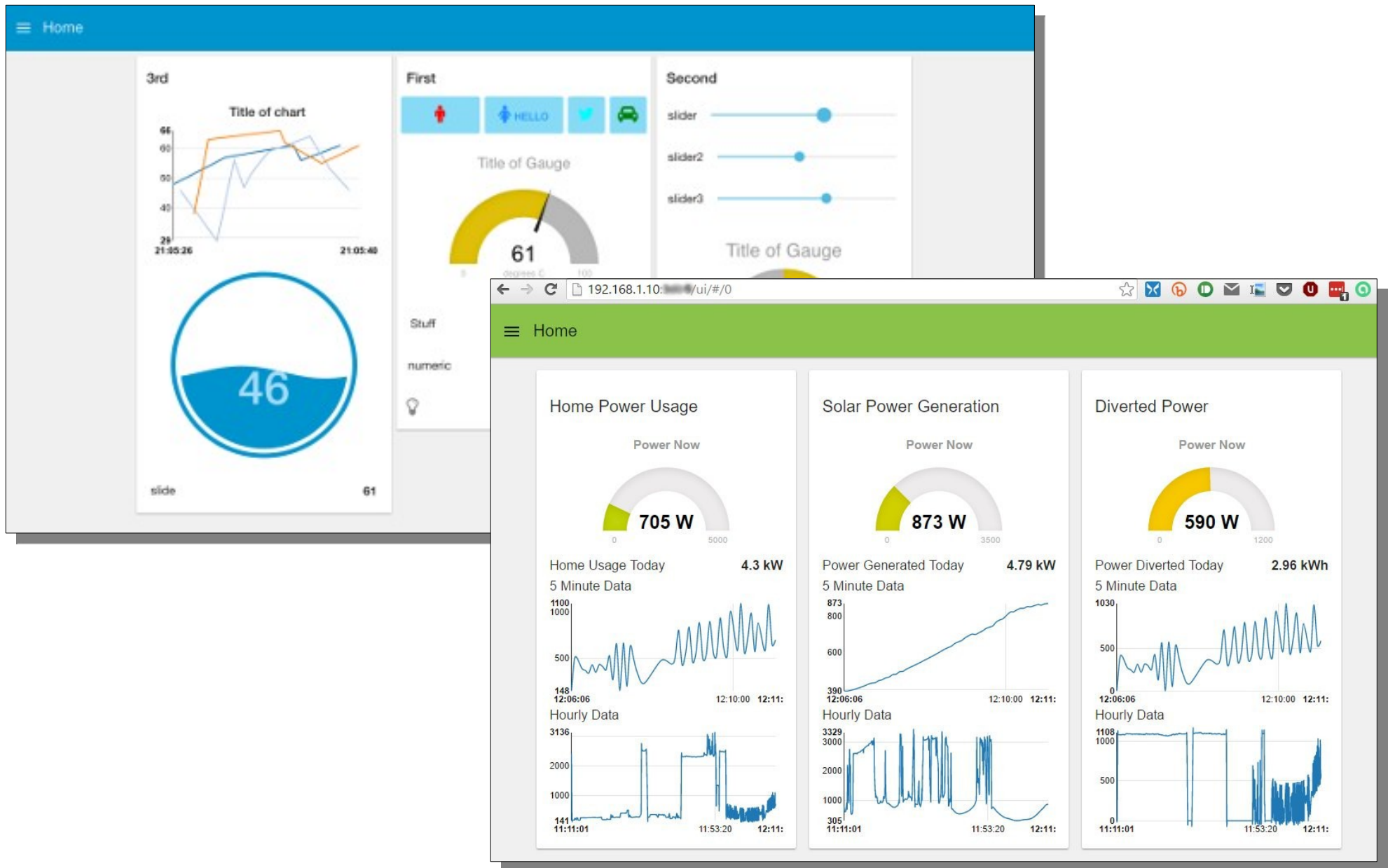
Note: This node does not send any response to the http request. This should be done with a subsequent HTTP Response node.

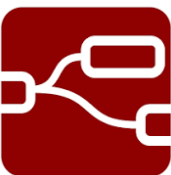






UI





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<https://www.raspberrypi.org/>



<https://flowfuse.com/>

Webinar - Develop, Manage, and Deploy Complex Node-RED Projects at Scale - [Register now](#) >



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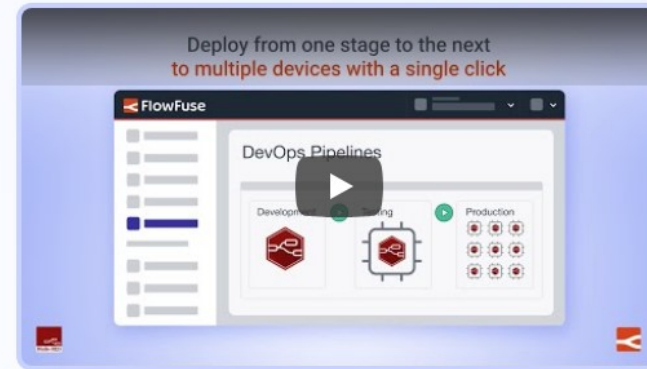
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Free

For edge device management. Free to you.

2 remote instances, and includes:

- ✓ 1 Team member
- ✓ Device auto snapshots

Free

[GET STARTED](#)

Starter

For smaller-scale applications. The fastest, easiest way to host Node-RED instances.

Starts at 2 hosted instances, 2 remote instances, and includes:

- ✓ 2 Team members
- ✓ Device management
- ✓ Device Auto Snapshots
- ✓ Blueprints
- ✓ Option to upgrade up to 3 hosted/3 remote instances

Starting at
\$20 /month
after free trial of Team

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Team

Empower your Node-RED developer teams with the collaborative features of FlowFuse Cloud.

Starts at 5 hosted instances or remote instances, and includes:

- ✓ 20 Team members
- ✓ Team collaboration
- ✓ MQTT Broker (8 clients to start)
- ✓ Standard Support + SLA: 99.5%

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\$175 /month

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Enterprise

For business critical applications that require high availability and enterprise support.

Starts at 20 hosted instances or remote instances, and includes:

- ✓ Unlimited team members
- ✓ Single Sign-On
- ✓ MQTT Broker (20 clients to start)
- ✓ Enterprise support + SLA: 99.9%
- ✓ Device group management

Let's Talk

[CONTACT US](#)



Recursos

git clone <https://gitlab.com/dgraselli/stack-nodered-mqtt>

Acceder a <http://localhost:1880>

Para UI acceder a <http://localhost:1880/ui>



Recursos

Comienza aquí

<https://nodered.org/docs/getting-started/>

Recetas simples

<https://cookbook.nodered.org/>

Nodos & Flujos

<https://flows.nodered.org/search?type=node>

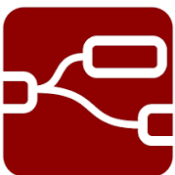
<https://flows.nodered.org/search?type=flow>

Node RED Programming Guide

<http://noderedguide.com/>

Youtube - Intro to Node-RED: Part 1 Fundamentals (1 de 5)

<https://www.youtube.com/watch?v=3AR432bguOY>



¡ Manos a la obra !